

PHYSICS

Courses

Below you will find a list of our current or recent offerings. See the [course catalog](#) for descriptions and updated information.

53-034 Introduction to Earth Science

This course examines the complex physical relations between land, sea and atmosphere. It also explores how some actions of our modern civilization disrupt the environment. Topics include geologic hazards, land management, water resources, hazardous waste disposal, energy resources, mineral resources and conservation of resources. Contributes to Environmental Studies. (NS) (Fall, Odd years)

53-064 Exploring the Universe: the Solar System

This course reviews our current scientific understanding of the constituents of our solar system: the sun, the terrestrial planets, the gas giant planets, the ice giants, dwarf planets, moons, rings, asteroids, comets, Kuiper belt objects, and the Oort cloud. Current theories of the origin and evolution of the solar system will be presented along with the history of its development and the basic science that governs the solar system. It will be discussed how new data from ground-based observatories, robotic space probes, and human space explorations continue to shape our theories of the solar system. Prerequisite: high school algebra.

Appropriate for first year students. (NS) (Fall)

53-074 Expl Universe: Stars, Galaxies, & Cosmo

This course reviews our current scientific understanding of the universe beyond our solar system. Topics include: stars, stellar structure, and stellar evolution including the formation of white dwarfs, neutron stars, and black holes. The structure of our Milky Way galaxy, other galaxies and theories of how they formed will be presented. Current theories of cosmology, the

robotic space probes continue to shape our theories of the universe. Prerequisite: high school algebra. Appropriate for first year students. (NS) (Spring)

53-104 Introduction to Engineering

Introduction to engineering as a discipline and a profession. Examines the design, manufacture, assembly and evaluation of products using individual and team projects.

Particular emphasis is placed on the role of written and graphical communication in the design process, the use of computer-aided design and rapid prototyping in product development and the role of entrepreneurship in high-tech ventures. Contributes to Design Thinking and Computer-Aided Design & 3D Printing Certificate. (NSL) (Spring)

53-112 Applications of CAD

Computer Aided Design software is used in various fields to create 2D and 3D designs, technical documentation, and product development. In this course students will learn how to use CAD software to create and manipulate 2D drawings (such as floor plans and elevations) and 3D models. Once created, the student will learn how to produce physical copies of these models using a laser cutter and/or a 3D printer. Contributes to Design Thinking and Computer-Aided Design & 3D Printing Certificate. (NS) (Fall)

53-154 Fundamentals of Physics I (3-3)

A calculus-based treatment of mechanics, wave motion, electromagnetism and optics.

Prerequisites: Concurrent registration or credit in Mathematics 52-164. (NSL) (Fall)

53-164 Fundamentals of Physics II (3-3)

A calculus-based treatment of mechanics, wave motion, electromagnetism and optics.

Prerequisites: Physics 53-154. (Spring) (NSL)

53-214 Modern Physics (3-3)

An introduction to thermodynamics and statistical mechanics, relativity, quantum mechanics, and the physics of atoms, nuclei and particles. Prerequisite: Physics 53-154, 53-164, Mathematics 52-164 and 52-264. Previous or concurrent registration in Mathematics 52-364, or consent of the instructor. (NSL) (Fall)

53-254 Observing the Universe

PHYSICS 53-104, or permission of the instructor. (NSL) (Fall, Even Years)

53-314 Engineering Design

This course introduces students to rapid prototyping. As part of this process students will deepen their proficiency in using Computer Aided Design software by creating complex single and multibody parts. They will also learn how to analyze the stresses on such parts. At the end of this course, students will have the opportunity to take the Autodesk Fusion Certified User exam. Contributes to Design Thinking and Computer-Aided Design & 3D Printing Certificate. Prerequisites: Physics 53-112 and 53-104. (NS) (Spring)

53-324 Electromagnetism I

An introduction to thermodynamics and statistical mechanics, relativity, quantum mechanics, and the physics of atoms, nuclei and particles. Prerequisite: Physics 53-154, 53-164, Mathematics 52-164, and 52-264. Mathematics 52-364 as a prerequisite or corequisite, or consent of the instructor. (NSL) (Fall)

53-334 Classical Mechanics I (3-3)

An advanced treatment of Newtonian mechanics with applications to kinematics, forced oscillations, central force motion and systems of particles. Prerequisites: Mathematics 52-364, Physics 53-154 and 53-164. Previous or concurrent registration in Mathematics 52-754. (NSL) (WA) (Spring)

53-364 Fund of Materials Science & Engineering

This course focuses on the emergence of structural properties from atomic and molecular-scale interactions by conducting a survey of three broad classes of materials: metals, ceramics and polymers. Particular attention will be paid to atomic structure and bonding, the structure of crystalline solids, phase diagrams, and the application and processing of polymers deployed in additive manufacturing (e.g., 3D Printing) and biomaterials. Also Chemistry 51-424. Contributes to Data Analytics and Data Science. Prerequisites: Physics 53-154. Chemistry 51-103 is a pre- or co-requisite for this course. (NS) (Spring)

53-404 Electronics (3-3)

53-413 Classical Mechanics II (3-0)

Introduction to the Lagrangian and Hamiltonian formulations of mechanics with application to non-inertial reference frames, rigid bodies and oscillating systems. Prerequisite: Physics 53-334. (NS) (Fall, Odd years)

53-424 Quantum Mechanics

A detailed introduction to quantum mechanics and its applications to atoms and molecules. Also Chemistry 51-714. Prerequisites: Physics 53-214 and 53-334. (NS) (Spring)

53-433 Electromagnetism II

Applications of Maxwells equations to propagation of plane and guided waves in various media. Prerequisite: Physics 53-324. (NS) (Spring, Odd years)

53-434 Statics

A study of forces, resultants and components of force systems, forces due to friction, conditions of equilibrium, forces acting on members of trusses and frame structures, centroids and moments of inertia. The lab portion of this course involves graphics and modeling fundamentals for engineering design: computer modeling of solid geometry and generation of engineering drawings. Introduction to computer programming using a high-level language and applications of computational methods to the solution of mechanical engineering problems. Contributes to Design Thinking. Prerequisite: Physics 53-154, Mathematics 52-164, and 52-264. Previous or concurrent registration in Mathematics 52-364 is required. (NS) (Fall).

53-454 Math Methods in Physical Science

This course examines select mathematical methods used in analyzing problems that arise in the physical sciences and engineering. Emphasis is given to both analytical and computer algebra or numerical approaches to problem solving and analysis. Analytical topics may include: linear vector spaces, Fourier series, ordinary differential equations, the calculus of variations, special functions, series solutions of differential equations, partial differential equations, functions of a complex variable, and integral transforms. Prerequisite: Mathematics 52-364, or permission of the instructor. Approved as an upper-level elective course for the Mathematics major,

53-463 Dynamics

A study of kinematics, force-mass-acceleration relations, work and energy, impulse and momentum as well as the moments of inertia of mass. Prerequisite: Physics 53-434, Mathematics 52-164, and 52-264. Previous or concurrent registration in Mathematics 52-364. (NS) (Spring, Odd years).

53-473 Thermodynamics

Basic principles of macroscopic thermodynamics including pressure-volume-temperature relationship of pure substances, work, heat, first and second laws of thermodynamics, entropy and the analysis of simple power and refrigeration systems. Prerequisite: Physics 53-154, 53-164, 53-214, Mathematics 52-164, 52-264, and 52-364. (NS) (Spring, Even years)

53-872 Capstone Course (2-0)

A scientific writing course for physics majors in which the capstone is drafted, discussed, revised and polished so that the writing is effective and conforms to modern standards of scientific journal style. In addition the student will learn how to fully integrate graphical elements and tables, mathematical equations, and numerical information into the document. The student and capstone course professor will work closely with the students capstone research advisor to bring the document to completion. Requirement: Students are expected to be finished with or nearing completion of the research phase of their capstone. (Spring)

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